

Assignment 1

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Optional Exercises

1. Uptime

```
ramisk@Ramis-Laptop:~/xv6-labs-2025$  
ramisk@Ramis-Laptop:~/xv6-labs-2025$  
ramisk@Ramis-Laptop:~/xv6-labs-2025$ make qemu  
mv -f fs.img fs.img.bk  
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_  
ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forp  
han user/_dorphan user/_sleep user/_sixfive user/_memdump user/_find user/_uptime user/_history user/_memdump  
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000  
ballocc: first 1199 blocks have been allocated  
ballocc: write bitmap block at sector 46  
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -  
drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
  
hart 1 starting  
hart 2 starting  
init: starting sh  
$ uptime  
16 ticks  
$ uptime  
54 ticks  
$ uptime  
71 ticks  
$ |
```

2.Support Regular Expressions

```
ramisk@Ramis-Laptop: ~/xv6  X + v  
init: starting sh  
$ find . a.*  
./cat  
./forphan  
./dorphan  
$ find . b.  
./zombie  
$ find . .*  
./README  
./findtest.sh  
./sixfive.txt  
./cat  
./echo  
./forktest  
./grep  
./init  
./kill  
./ln  
./ls  
./mkdir  
./rm  
./sh  
./stressfs  
./usertests  
./grind  
./wc  
./zombie  
./logstress  
./forphan  
./dorphan  
./sleep  
./sixfive  
./memdump
```

3. Modify the shell to support wait

```
ramisk@Ramis-Laptop:~/xv6-labs-2025$  
ramisk@Ramis-Laptop:~/xv6-labs-2025$ make qemu  
mv -f fs.img fs.img.bk  
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_  
ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forp  
han user/_dorphan user/_sleep user/_sixfive user/_memdump user/_find user/_uptime user/_history user/_memdump  
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000  
ballocc: first 1198 blocks have been allocated  
ballocc: write bitmap block at sector 46  
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -  
drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
  
hart 1 starting  
hart 2 starting  
init: starting sh  
$ sleep 40 &  
$ wait  
$ echo done  
done  
$ |
```

4. modify the shell to support tab completion

```
ramisk@Ramis-Laptop:~/xv6-labs-2025$  
ramisk@Ramis-Laptop:~/xv6-labs-2025$ make qemu  
mv -f fs.img fs.img.bk  
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_  
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drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
  
hart 1 starting  
hart 2 starting  
init: starting sh  
$ testing ec then tab  
exec testing failed  
$ echo  
$ testing he then tab  
exec testing failed  
$ he  
exec he failed  
$ testing us then tab  
exec testing failed  
$ usertests  
usertests starting  
$ |
```

5. modify the shell to not print a \$ when processing shell commands from a file

```
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mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_  
ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forp  
han user/_dorphan user/_sleep user/_sixfive user/_memdump user/_find user/_uptime user/_history user/_memdump  
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drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0  
  
xv6 kernel is booting  
  
hart 1 starting  
hart 2 starting  
init: starting sh  
$ sh < sixfive.txt  
exec 5 failed  
exec 3 failed  
exec 127 failed  
exec 100 failed  
exec 18-4 failed  
exec 06 failed  
$ |
```

6. modify the shell to keep a history of passed shell commands

```
ramisk@Ramis-Laptop:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_
ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forp
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drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ echo hi
hi
$ echo bye
bye
$ history
echo hi
echo bye
history
$ |
```